J&A AUTOMATION

Technical report

Template 2026.09.02.2

CP020 · BBS Mexico · Aug 5, 2026

Operational record

Project	CP020 · BBS Mexico
Client	IMPC Gmbh
Date	Aug 5, 2026
System	Main Assembly Cell PLC
Site	BBS Saltillo Plant
Area / line	Assembly Line 2
Station / machine	Station 10 - 14 Transfer
Status	Approved
Safety-related	Yes

Technical records

System type	PLC & Safety Controller
PLC platform	Siemens S7-1500 & Allen Bradley GuardLogix
Controller	CPU 1517F-3 PN/DP
HMI / SCADA	Siemens WinCC Unified & FactoryTalk View

Network protocol

PROFINET / EtherNet/IP Industrial Gateway

Software version

TIA Portal v19 / Studio 5000 v35

Program reference

BBS_MEX_L2_MAIN_REV_3.4

Change summary

Problem / symptom

Intermittent emergency stop trip on conveyor index transfer

Diagnosis / root cause

Dual-channel safety relay pulse test timing discrepancy between safety gate and PLC safety input module

Change performed

Adjusted discrepancy time window in Safety Evaluation configuration from 50ms to 200ms per manufacturer recommendation

Production impact

Zero unplanned downtime observed over 4 hours of test production

Validation

Tested 10 consecutive emergency stop triggers and re-engaged safety circuit; all fault codes reset cleanly

Validation result

Pass · Fully functional and within safety SIL3 compliance standards

Open risk

None identified; hardware and firmware signatures backed up

Rollback plan

Restore safety project file BBS_MEX_L2_MAIN_REV_3.3 from local backup PLC SD card

Technical changes

DATE	DETAIL	CHANGE SUMMARY	STATUS
Aug 5, 2026	Safety Input Module F-DI 16x24VDC	Updated input filter parameter and discrepancy time window	Approved
Aug 5, 2026	HMI Alarm Banner	Added explicit diagnostic text: Station 12 Gate Discrepancy Fault	Approved